MIL-S-22432/32C
22 August 1974
SUPERSEDING
MIL-S-22432/32B
19 MARCH 1968
MIL-S-22432/32A(WP)
21 APRIL 1965
MIL-S-22432/32(WP)
9 MARCH 1965

MILITARY SPECIFICATION SHEET
SERVOMOTOR, TYPE 15SM4d

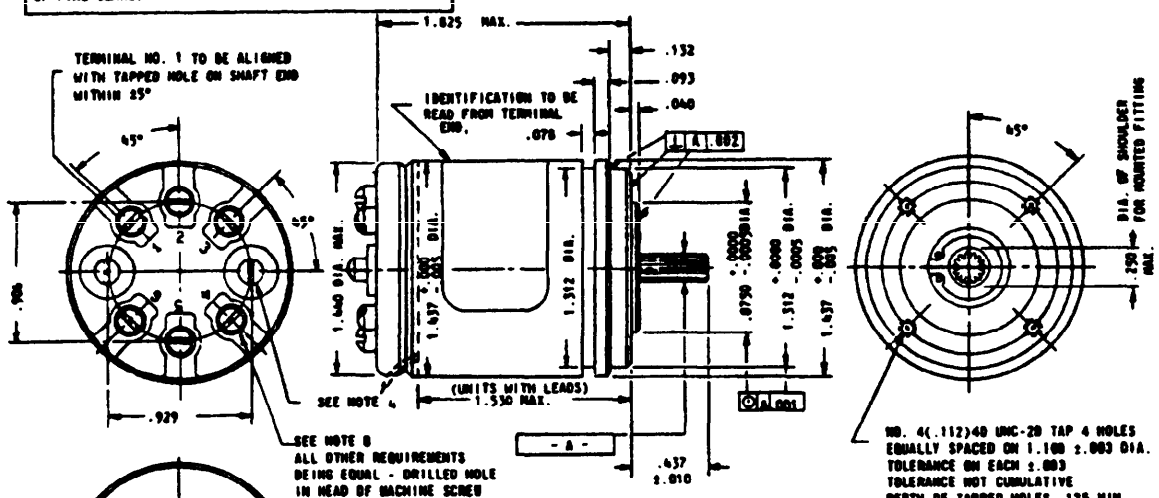
This Specification is Mandatory for Use by All Departments and Agencies of the Department of Defense.

THE COMPLETE REQUIREMENTS FOR PROCURING THE SERVOMOTOR DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF SPECIFICATION MIL-S-22452.

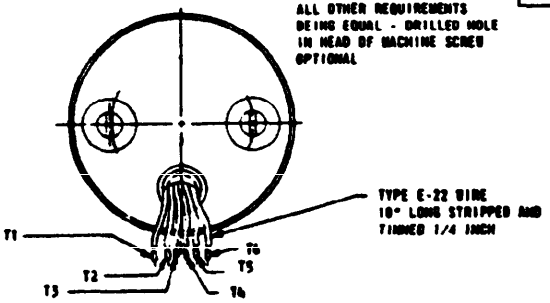
SPUR GEAR DATA	
TEETH	15
PITCH	64
OUTSIDE DIAMETER	.1770 +.0000 -.0010
PRESSURE ANGLE	20°
TOOTH FORM	FULL DEPTH INVOLUTE

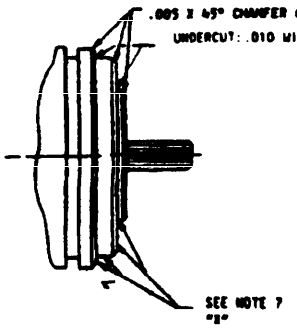
DESIGN TO AGMA 207.04(ASA B6.7-1956) "20 DEGREE INVOLUTE FINE PITCH SYSTEM" PRECISION GRADE 1
TESTED TO AGMA 236.04(ASA B 6.11-1956) INSPECTION OF FINE GEARS.

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES ON FRACTIONS ±1/64
DECIMALS ±.005 ANGLES ±2°



NO. 4(.112)40 UNC-2B TAP 4 HOLES EQUALLY SPACED ON 1.100 ±.003 DIA. TOLERANCE ON EACH ±.003 TOLERANCE NOT CUMULATIVE DEPTH OF TAPPED HOLES .125 MIN. NOTE: BOTTOM OF HOLES SHALL BE CLOSED.





NOTES:

1. PERPENDICULARITY AND CONCENTRICITY SHALL BE MEASURED WITH THE UNIT IN A VERTICAL POSITION SUPPORTED BY THE SHAFT. THEN THE HOUSING SHALL BE ROTATED.
2. RUNOUT OF FEATURE 'A' SHALL NOT EXCEED .0014 TIR.
3. END PLAY AND RADIAL PLAY SHALL BE MEASURED DURING THE REVERSAL OF THE FOLLOWING LOADS:
   END PLAY - 1 POUND RADIAL PLAY - 1/2 POUND
4. THE DESIGN OF THIS UNIT SHALL PROVIDE FOR THE TERMINAL BLOCK TO REMAIN FIXED WHEN TERMINAL SCREWS ARE REMOVED.
5. THE END CAP DESIGN SHALL BE SUCH THAT THE TERMINAL LUG MAY BE INSTALLED WITH THE BARREL INBOARD OR 180° POSITION OUTBOARD. RECESSES FOR LUGS SHALL BE OF WIDTH AND DEPTH TO PREVENT TURNING OF THE LUGS.
6. THE FOLLOWING LIST OF HARDWARE MS-35270-213 MACHINE SCREW (6 REQUIRED), MS-35338-135 LOCK WASHER (6 REQUIRED) SHALL BE PACKAGED IN A VENTILATED ENVELOPE MADE OF MATERIAL HAVING A PH VALUE OF 6-8 THAT IS COMPATIBLE WITH THE WEIGHT OF THE CONTENTS. THEN THE PACKAGED HARDWARE SHALL BE PLACED IN THE UNIT SERVOMOTOR PACKAGE.
7. HOUSING SHALL BE STAINLESS STEEL. SURFACES MARKED "X" SHALL BE BRIGHT METALLIC FINISH (AS PER MIL-C-13924).
8. DIMENSIONS AND TOLERANCES COMPLY WITH ANSI Y14.5.
9. PENDING ISSUANCE OF A QUALIFIED PRODUCTS LIST (QPL) A SAMPLE OF THE SERVOMOTORS TO BE FURNISHED UNDER THIS SPECIFICATION SHALL BE SUBJECTED TO THE PREPRODUCTION TESTS AS SPECIFIED IN SPECIFICATION MIL-S-22432.

FSC 6105

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:16Z
Check the source to verify that this is the current version before use.

TABLE I
SERVOMOTOR, TYPE 15SM4d

Performance Requirements and Characteristics

Requirements	Value	Unit	Tolerance
Electrical:			
Frequency:			
Rated	400	hz	
Test	400	hz	±1%
Range	380 to 420	hz	
Voltage:			
Reference phase, rated	115	volts rms	
Reference phase, test	115	volts rms	±1%
Reference phase, range	103.5 to 126.5	volts rms	
Control phase, rated (series)	36	volts rms	
Control phase, test (series)	36	volts rms	±1%
Control phase, range (series)	0 to 40	volts rms	
Control phase, rated (parallel)	18	volts rms	
Control phase, range (parallel)	0 to 20	volts rms	
Current:			
Reference phase	.117	amperes rms	maximum
Control phase (series)	.368	amperes rms	maximum
Control phase (parallel)	.680	amperes rms	nominal
Power:			
Reference phase	6.8	watts	maximum
Control phase (series)	6.8	watts	maximum
Control phase (parallel)	6.0	watts	nominal
Synchronous Speed	8000	rpm	
Impedance:			
Reference phase	526+j935	ohms	nominal
Control phase (series)	52+j94	ohms	nominal
Power Factor:			
Reference phase	.49		nominal
Control phase (series)	.49		nominal
Effective Resistance:			
Reference phase	2200	ohms	nominal
Control phase (series)	216	ohms	nominal
Control phase (parallel)	54	ohms	nominal
Minimum Starting Voltage	.75	volts rms	maximum
Coupling in Control Winding	.8	ratio	minimum
Mechanical:			
No-load Speed	6200	rpm	minimum
Stall Torque			
at 23°C	1.2	ounce-inches	minimum
at 125°C	.90	ounce-inches	minimum

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:16Z
Check the source to verify that this is the current version before use.

TABLE I (Continued)

Requirements	Value	Unit	Tolerance
Rotor Moment of Inertia	.8	gm-cm^2	maximum
Radial Play	.0008	inch	maximum
End Play	.0035	inch	maximum
	.0010	inch	minimum
Temperature Rise	65	degrees C	maximum
Noise, Mechanical	not excessive		
Time Constant	.006	seconds	nominal
Reversing Time	.010	seconds	nominal
Theoretical Acceleration at Stall	105000	radians/second^2	nominal
Maximum Power Output	.8	watts	nominal
Weight	7.3	ounces	maximum

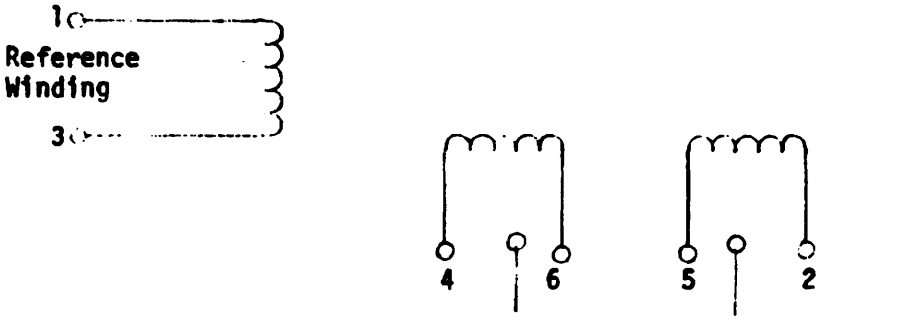
Notes:

1. The servomotor identified herein shall conform to all requirements, and shall be subjected to all tests delineated in MIL-S-22432, with the following exceptions:

Salt Atmosphere Resistance, paragraph 3.30
Audible Noise, paragraph 3.31

2. Type 15SM4d servomotors shall be capable of operation in ambient temperatures ranging from -55°C to 125°C.

3. Connection Diagram:



Control Winding Excitation 400 cps

Figure 2

a. For CCW rotation the voltage at terminal 2 leads the voltage at terminal 1 by 90 electrical degrees.

b. For series operation of control winding terminals 5 and 6 are strapped and 36 volts rated is applied across terminals 2 and 4.

c. For multiple or parallel operation of control winding terminals 6 and 2 are strapped, and terminals 4 and 5 are strapped. 18 volts rated is applied across terminals 2 and 4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:16Z
Check the source to verify that this is the current version before use.

4. Sampling, inspection and preparation for delivery shall be in accordance with MIL-S-22432.

Part Number: M22432/32 plus the appropriate dash Number from Table II.

TABLE II

Dash No.	Shaft	Termination
1	Spline	Board
2	Spline	Leads

Custodians:

Army - MU
Navy - AS
Air Force - 11

Review Activities:

Army - EL, MI, MU
Navy - SH, AS, OS, EC
Air Force - 11, 82

User Activities:

Army - ME, AV
Navy - OS, MC, CG

Preparing Activity:

Navy - AS

DOD Project Number:

6105-0112

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:16Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-S-22432/32C SERVOMOTOR, TYPE 15SM4d

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:16Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Systems Command
Washington, D.C. 20361

POSTAGE AND FEES PAID
DOD 316

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NAVAL AIR SYSTEMS COMMAND(AIR-52021)
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20361

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:16Z
Check the source to verify that this is the current version before use.